

Hirav Thakur

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Computer Science, Honours with Co-operative Program, Artificial Intelligence Specialization, History Minor
September 2022 - April 2027

EXPERIENCE

Ford Motor Company (Ford Pro)

Waterloo, ON

Full-Stack Software Engineering Intern

September 2026 - December 2026

Wilfrid Laurier University

Waterloo, ON

Undergraduate Researcher - Vision Language Models & Robotics

August 2026 - Present

- Conducting research on **Vision Language Model (VLM) - assisted indoor robot navigation**, investigating methods for enabling autonomous robots to reason about and interact with obstacles in dynamic environments.
- Evaluating VLM\LLM based navigation approaches using **ROS2 and Nav2** with a focus on interactable-object detection, active perception, and navigation recovery.
- Investigating open-research problems in **dynamic scene adaptation and long-term spatial memory** for autonomous indoor navigation.

Cardea Health

Mississauga, ON

NLP/OCR & Data Engineering Intern

May 2025 - December 2025

- Led schema standardization across **20–40+ views per clinical trial**, enabling LLM-driven (Gemini) query generation and automated patient eligibility analysis over **100K+ clinical documents**.
- Contributed to the design of a Common Data Model (CDM) supporting data unification across **5 client EMR systems** (Canada & US).
- Designed and scaled AWS data pipelines using **Python** and **SQL (Athena)** to process datasets ranging from **3K to 1.2M+ patient records**, supporting on-demand and triggered workflows.
- Built NLP pipelines (MedSpaCy) in **Python** for PHI detection and entity resolution, processing **100K+ documents**.
- Reduced clinical data processing time by **50%**, accelerating patient selection for trials.
- Engineered dynamic data refresh and deployment systems, replacing partially manual workflows.
- Improved matching accuracy and pipeline reliability by unifying fragmented EMR schemas.
- Supported clinical trial workflows (**2–3 trials/week**) by delivering structured datasets for eligibility analysis and dashboards.

Propel Holdings

Toronto, ON

Software Development Intern

September 2024 - December 2024

- Migrated **80+ APIs (30% of system)** to a Redux-based architecture using **React**, improving scalability and reducing bugs.
- Reduced page load times from **5s to 1s (5× faster)** by optimizing **React-based** frontend data flow and API integration.
- Developed features for a **React-based** loan management platform used by **50+ internal agents**, improving usability and workflow efficiency.
- Optimized backend SQL queries across **3+ internal tools**, achieving **2× faster** performance.
- Resolved production issues related to formatting and business logic inconsistencies.
- Delivered **10–15 tickets per sprint** in a 2-week Agile cycle (JIRA).

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, SQL, C++, C, Java, HTML/CSS, Swift, Bash, PowerShell, Golang, Kotlin, Scala

Technologies: AWS (S3, Glue, Athena), PyTorch, HuggingFace, OpenAI API, FastAPI, React Flow, Azure, GCP, Git, REACT.js, React Native, MongoDB, Kubernetes, Apache Camel, GCP, Docker, Terraform, Kafka, IBM MQ, Supabase, PostgreSQL, MySQL, Django Springboot, Redux, .NET, Node.js/Express, Unix/Linux, SQLite, LINQ

Certifications: Microsoft Certified: Azure AI Fundamentals, Microsoft Certified: Azure Fundamentals, Transformers Models and BERT Models by Google Cloud